Returns to Representation: Girls' Youth Sports Participation Following the Founding of the WNBA

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Abstract

This paper investigates the impact of a women's professional sports league on girls' youth sports participation. Using the Women's National Basketball Association's (WNBA) inaugural season in 1997 as treatment, I apply the Callaway and Sant'Anna [2020] estimator to determine if there are any effects of the league's formation on the participation of girls in high school sports. Across four sports, I find a positive, though imprecisely estimated, effect on per capita basketball participation, while track and field, cross-country, and soccer do not exhibit any statistically significant effects.

1 Introduction

The Women's National Basketball Association (WNBA) was founded in 1997 and is the longest-running women's professional sports league in the United States. The league has grown significantly since its inception, with 15 teams currently competing in the league and further expansion on the horizon. The WNBA, in recent years, has become a prominent platform for the promotion of women's sports and has seen record viewership and attendance [WNBA, 2024]. The presence of a sports team has potentially far-reaching effects on the local community, including increased interest in the sport, improved facilities, and increased opportunities for youth participation. However, the extent to which the presence of a WNBA team has influenced youth sports participation has not been rigorously studied. This is a particularly important question given previous research on the effects of Title IX, which found that increased opportunities for girls in sports led to significant increases in sports participation and subsequent increases in college enrollment and labor force participation for women Stevenson [2010].

I aim to fill this gap in the literature by examining the impact of the WNBA's formation on girls' high school sports participation. Using a differences-in-differences approach, I compare the changes in sports participation in states that received a WNBA team to those that did not. The staggered rollout of WNBA teams across states allows for variation in treatment timing, and the presence of never-treated states provides a natural control group for comparison. The main findings are largely null: while point estimates for basketball are positive, no sport exhibits effects that are robust to permutation-based inference, and the results are sensitive to the choice of outcome scaling. This null result is itself informative, suggesting that any effects of WNBA franchise entry on girls' sports participation are not detectable at the state level with available data.

The rest of the paper is organized as follows. Section 2 reviews the relevant literature on the effects of professional sports leagues on youth sports participation. Section 3 describes the data. Section 4 introduces the estimation strategy. Section 5 presents and discusses the main results, while Section 6 discusses robustness findings. Finally, Section 7 concludes.

2 Literature

The literature on girls' sports participation follows a few main threads. First, Stevenson [2010] was an important paper in the field, as it was one of the first to use a differences-in-differences strategy to estimate the effect of Title IX on girls' sports participation. The paper finds that Title IX had a significant effect on sports participation and subsequent increases in college enrollment and labor force participation for women. Beck and LaLumia [2022] is a more recent paper which is similar in spirit to this paper, but differs in application. Studying the All-American Girls Profession Baseball League (AAGPBL), the paper finds that the presence of a professional team raised female labor force participation and employment in the treated cities. Reilly and Rodriguez [2024] is the paper I build upon. The paper studies the same question as this one, but uses standard two-way fixed effects methods, which can lead to biased estimates in the presence of heterogeneous treatment effects across cohorts and over time. This paper uses more modern methods to address this concern.

There is also literature on the role model effect of professional athletes and teams on youth sports participation, such as Frick and Wicker [2016], which finds that the success of the German national football team increased ameteur football participation in Germany. Feddersen and Maennig [2009] more directly looked at the effect of individual athletes, finding that while participation grew as German players ascended, they find deterioration as tennis players like Boris Becker and Steffi Graf retired. My paper fits nicely into this literature, as it looks at the effect of a professional sports league, rather than individual athletes, on youth sports participation.

In the econometric literature, this paper heavily employs the methods of Callaway and Sant'Anna [2020], which is a staggered differences-in-differences estimator that allows for heterogeneous treatment effects across cohorts and over time. This is particularly useful in this context, as the staggered rollout of WNBA teams across states allows for variation in treatment timing, and the presence of never-treated states provides a natural control group for comparison. Under the findings of Goodman-Bacon [2021] and De Chaisemartin and D'Haultfœuille [2020], we know that the standard two-way fixed effects model as used in Reilly and Rodriguez [2024] is biased in a staggered treatment environment. I also apply the synthetic differences-in-differences method of Arkhangelsky et al. [2021] as a robustness check, which constructs a synthetic control group for each treated unit based on a weighted average of the characteristics of the control units.

3 Data

National Federation of State High School Associations (NFHS) Data

The primary data source for this analysis is National Federation of State High School Associations (NFHS) data on high school sports participation. The NFHS collects data on the number of participants in various sports at the high school level across the United States. I focus on four sports: basketball, track and field, cross-country, and soccer. These were chosen due to track and cross-country's season not directly competing with basketball, while soccer is a direct competitor for participants. They are also very popular sports and had consistent data availability across the time period studied.

Population level data is normalized by state population, which is obtained from the US Census Bureau. This is done to account for differences in state population sizes, which can influence the number of participants in high school sports. Normalizing by population allows for a more accurate comparison of sports participation across states and over time. The main results are presented as increase in participation per 10,000 people, which is a more interpretable metric than raw participation numbers, especially when comparing across states of varying sizes.

Data from NFHS spans 1993 to 2016, allowing for a pre-treatment period of four years and a post-treatment period of 6 years. The data is aggregated at the state-year level, and I balance the panel so that each state has observations for all year in the sample. The final dataset includes 47 states and 24 years, results in a total of 1,128 state-year observations. Table 1 provides summary statistics for the key variables in the analysis for the treated and never-treated states. We can see here that never-treated states have higher per 10,000 population participation rates across all sports, which is likely a result of the fact that the treated states are generally more populous, and thus have more participants in high school sports, but when normalized by population, the treated states have lower participation rates. The difference is largest in basketball, with a mean difference of approximately six participants per 10,000 people, while the difference is smallest for cross-country, with a mean difference of approximately 0.3 participants per 10,000 people. The differences in participation rates across treatment status highlight the importance of controlling for population size in the analysis, as larger states are more likely to receive a WNBA team and also have higher participation rates in high school sports. Table 2 provides summary statistics for the treated states by cohort, where a cohort is defined as the year in which a state receives its first WNBA team. This table highlights that, for the majority of cohorts, mean participation rates are largely stable across cohorts. Later cohorts, which are generally smaller and have fewer treated states, have lower participation rates across all sports.

WNBA Data

The treatment variable in this analysis is the presence of a WNBA team in a given state, specifically the first year in which a game is played. If a team is ever active in that state, then the state is considered treated in perpetuity following the year of the team's introduction. Each state that receives a WNBA team is assigned to a cohort based on the year in which the team was introduced. The WNBA data is collected from the league's official website, which provides information on the location and entry year of each franchise. Figure 1 shows the locations of WNBA franchises by state and entry year. The WNBA had a staggered rollout, with the first teams introduced in 1997 and additional teams introduced in subsequent years, with the most recent team studied introduced in 2010. The staggered nature of the treatment allows for a clean estimation of the average treatment effect on the treated (ATT) using the methods of Callaway and Sant'Anna [2020], which is robust to heterogeneous treatment effects across cohorts and over time.

Limitations of the Data

There are some limitations and issues with the data that need to be noted. Firstly, in 2011 the NFHS changed the way they collected and reported participation data, which biases the data upwards slightly. This is a minor concern as in the framework of Callaway and Sant'Anna [2020], this is absorbed as a year fixed effect. It was also implemented to impact both never-treated and treated states equally. Thus, we should have no concern surrounding differential reporting. Secondly, Pennsylvania cross-country individually changed its reporting in 2008. This is also worth mentioning but not of much concern as our estimation drops Pennsylvania from the analysis due to the fact it is not balanced. Finally, the NFHS data is only at the state level, which limits the precision of the exercise. Ideally, we would have data more suitable to city level treatment, but this is not available. There is a concern, then, that larger states with a WNBA team may have a more muted effect since the treatment is only in one city, but this is a limitation of the data that cannot be overcome. Lastly, the decennial census interpolation has the potential to introduce measurement error into the population data, especially for fast growing states. However, since this normalization is applied to both treated and control states, and the interpolation is likely to be reasonably accurate at the state level, this is not a major concern for the analysis.

Balancing the panel in a sample like this introduces a tradeoff: we lose some treated units, but we gain the ability to use a more flexible estimation strategy that allows for heterogeneous treatment effects across cohorts and over time. I am comfortable with this tradeoff, as the Callaway and Sant'Anna [2020] estimator is robust to heterogeneous treatment effects, while a two-way fixed effects estimator would not be. We do lose one period of cross-country for a treated unit, California, which is initially concerning. But, California is in the first cohort, which is the largest cohort with seven treated states, so the loss of one treated unit is not as concerning as it would be for a smaller cohort. Further, it is a year post-treatment for California, so the loss of this is not as concerning as it would be if it were a pre-treatment period, which would be more informative for the parallel trends assumption.

4 Estimation

Using a differences-in-differences framework, I estimate the effect of the presence of a WNBA team on girls' sports participation. The staggered rollout of WNBA teams across states allows for variation in treatment timing, and the presence of never-treated states provides a natural control group for comparison. Following Callaway and Sant'Anna [2020], I estimate the following:

$$\hat{ATT}_{gt} = (Y_t - Y_{g-1}|G = g) - (Y_t - Y_{g-1}|\text{WNBA} = 0) \tag{1}$$

where Y_t is the participation in a given sport in year t for a state which receives a WNBA team in treatment cohort g, and Y_{g-1} is the participation in the year prior to treatment for that cohort. The second term is the same difference but for states that do not ever receive a WNBA team. I then aggregate these estimates across cohorts to obtain an overall estimate of the effect of having a WNBA team. Standard errors are clustered at the state level to account for serial correlation within states over time.

This estimator yields group-time average treatment effects, which allows for flexibility in the timing of treatment and does not require the parallel trends to hold for all groups and time periods, but only for the specific group-time pairs being compared. This is particularly useful in this context, as the staggered rollout of WNBA teams across states allows for variation in treatment timing, and the presence of never-treated states provides a natural control group for comparison. I use the simple weighting scheme proposed by Callaway and Sant'Anna [2020], which weights each group-time pair by cohort size, to aggregate the estimates across cohorts and obtain an overall estimate of the effect of having a WNBA team on girls' sports participation.

For this estimation strategy to be valid, two key assumptions must hold. First, the parallel trends assumption needs to hold, which states that, absent treatment, the treated and control states would have followed similar trends in sports participation. I provide evidence in support of this assumption in the results section. Second, there cannot be anticipation of treatment. That is, girls' sports participation should not have increased in anticipation of a preeminent WNBA team coming to their state. Given the nature of the treatment, I am comfortable assuming this is not a concern, but is also shown to be satisfied in the event study results.

Further, I note that the staggered nature of the treatment and the presence of never-treated states allows for a clean estimation of the ATT which is robust to heterogeneous treatment effects across cohorts and over time. It is worth discussing why I abstain from using a two-way fixed effects estimator in this context. A two-way fixed effects estimator would use units which are already treated as controls for units which are treated later, which can lead to biased estimates if there are heterogeneous treatment effects across cohorts and over time. The Callaway and Sant'Anna [2020] method is thus preferred for reasons mentioned above.

I also employ numerous robustness checks. These include the synthetic differences-in-differences method of Arkhangelsky et al. [2021], which constructs a synthetic control group for each treated unit based on a weighted average of the characteristics of the control units, the placebo permutation test in which I randomly assign treatment to states and re-estimate the main results 500 times to obtain a distribution of placebo treatment effects, and a leave-one-out check in which I drop the three most populous states in the sample, which are all in the first cohort, and re-estimate the main results.

5 Results

Figure 3 shows the event study estimates for basketball participation. The estimates support the parallel trends assumption if taken at face value, but there are massive confidence intervals in periods both pre and post-treatment. However, the results present a clear increase in basketball participation following treatment, with an estimated increase of two to three participants per 10,000 people. The results for track and field (figure 4) show a slight increase in participation, though statistical significance is borderline. For cross-country (figure 5) and soccer (figure 6) results are less clear, with mostly nulls and large confidence intervals.

When looking at results by cohort, we see that the effect is not uniform across cohorts. For basketball (figure 7), the effect is largest for the second cohort, in which two states received WNBA teams in 1998, with an estimated increase of nine participants per 10,000 across the states. The effect of a team being in a state appears quite random, with cohort 3 (1999) having a massive confidence interval, and cohorts 4-8 presenting increases of anywhere between zero and three participants per 10,000.

The results for track and field (figure 8) are similar to the basketball estimates across cohorts, with all but one cohort exhibiting increases in participation following treatment, though the confidence intervals are quite large. Only cohorts 2 (1998), 5 (2003), 7 (2008), and 8 (2010) have confidence intervals that do not include zero, with effects ranging from a decrease of one participant per 10,000 people to an increase of about 5 participants per 10,000 people. The other cohorts have confidence intervals that include zero, with point estimates that range from a decrease of one participant per 10,000 people to an increase of two participants per 10,000 people.

The results for cross-country are presented in figure 9. The results are even less promising for cross-country, with only one cohort exhibiting a statistically significant increase in participation following treatment, which is cohort 5 (2003), with an estimated increase of about just under one participant per 10,000 people. The other cohorts have confidence intervals that include zero, with point estimates that range from a decrease of under one participant per 10,000 people to an increase of about one participant per 10,000 people.

The results shown in figure 10 show that the effect is also largely null for soccer. Only cohorts 7 (2008) and 8 (2010) experience any effect discernible from zero with increases of just over two participants per 10,000 each. Every other cohort exhibits patterns ranging from decreases of one and a half participants per 10,000 to nearly three.

Overall, it is hard to draw strong conclusions from the results, as the confidence intervals are very large and there is significant hetereogeneity across cohorts. However, the point estimates are generally in the expected direction, with increases in participation following treatment, though the magnitudes are somewhat smaller than expected. However, the results are not significant at conventional levels, and the confidence intervals are quite large, which suggests that the results are not robust to inference and should be interpreted with caution. The results for basketball are the most informative, with an estimated increase of about two participants per 10,000 people following treatment, but even this result is not statistically significant at conventional levels.

6 Robustness

The results above present a view that the presence of a WNBA team in a given state is at lease a null effect and at most a potential increase in girls' sports participation. However, there are some concerns with the estimation strategy that need to be addressed. For instance, it is fair to assume that parallel trends may not hold for Callaway and Sant'Anna [2020]. To address this concern, I apply the synthetic differences-in-differences method of Arkhangelsky et al. [2021], which constructs a synthetic control group for each treated unit based on a weighted average of the characteristics of the control units. Here, the synthetic control group is constructed using the pre-treatment characteristics of the treated states, and the post-treatment outcomes of the synthetic control group are used to estimate the counterfactual outcomes for the treated states in the absence of treatment. This method allows for a more flexible estimation strategy that does not rely as heavily on the parallel trends assumption, and can provide more robust estimates of the treatment effect. Assumptions needed are stable unit treatment value assumption (SUTVA), which states that the potential outcomes for any unit are unaffected by the treatment status of other units, and that there are no spillover effects between treated and control units. This is a reasonable assumption in this context, as the presence of a WNBA team in one state is unlikely to systematically affect sports participation in another state. Additionally, the method assumes that the synthetic control group is a good approximation of the counterfactual outcomes for the treated states in the absence of treatment.

The results from this method are presented in figures 11 through 14, and show similar patterns to the main results, with increases in participation following treatment for basketball, though the magnitudes are somewhat smaller. For figure 11, the estimated increase in basketball participation following treatment is about two participants per 10,000 population, which is somewhat smaller than the main estimate of about two and a half participants per 10,000. For track and field (figure 12), cross-country (figure 13), and soccer (figure 14), I return a null result, with a mean confidence interval that includes zero and point estimates that are indistinguishable from zero[1]. This shows that the main results are not uniquely driven by the choice of estimation strategy, and that the main results are robust to a more flexible estimation strategy that does not rely as heavily on the parallel trends assumption.

The results of the permutation placebo test (figures 15 - 18) show that the actual estimated treatment effect is not statistically different than 95% of the placebo treatment effects for all sports. This drives home the point that the results are not stable and are not robust to inference, which is likely a result of the small number of treated units in each cohort, which leads to imprecision in the estimates.

I also perform a leave-one-out check, in which I drop the three most populous states in the sample, which are all in the first cohort, and re-estimate the main results. The results of this robustness check are presented in figure 19, and show that, for basketball, dropping Texas actually very slightly improves percision, while dropping California and New York results in very similar estimates to the baseline. For the other sports, the results are much in line with the baseline, with some minor changes in the point estimates and confidence intervals. These findings suggest that, while the first cohort is the largest and has the most treated units, the results are likely not being driven by any one state, and that the baseline findings are largely robust to dropping any one of the three most populous states in the sample.

7 Discussion

When comparing results across estimators (Table 3), the three approaches tell a consistent story. Basketball is the only sport for which all three estimators return a positive point estimate, with ATTs ranging from 2.1 to 2.2 participants per 10,000 population. For soccer, cross country, and track and field, point estimates are small and in some cases negative across all three estimators, providing little evidence of any effect. The consistency across TWFE, Callaway-Sant'Anna, and synthetic DiD is reassuring in one sense, implying that the null result is not an artifact of any particular estimation choice. However, the Callaway-Sant'Anna estimator carries larger standard errors than the other two, reflecting its flexibility in accommodating heterogeneous treatment effects at the cost of precision. With a maximum of seven treated states in any cohort, this imprecision is a fundamental constraint of the data rather than a methodological failure. Even for basketball, where the point estimate is economically suggestive, the permutation p-value of 0.518 indicates the result is indistinguishable from chance under the null. The combined evidence presented suggests a null result under the current research design using state-level data.

[1]Note: for cohorts of size $N = 1$, the synthetic differences-in-differences was implemented to recover placebo standard errors instead of bootstrapped standard errors as in larger cohorts.

8 Conclusion

Using population-normalized outcomes and permutation-based inference, I find no sport exhibits effects robust to the null hypothesis of no treatment. Basketball returns a consistent positive point estimate of approximately 2.2 participants per 10,000 across all three estimators, but these findings are not statistically distinguishable from chance. The results suggest that any effect of WNBA franchise entry on girls' sports participation is not detectable at the state level, a finding that is itself informative, as it suggests prior work reporting significant effects in levels may have been capturing population size rather than genuine behavioral change.

The primary issue with the results are the large confidence intervals. This is likely a result of the small number of treated units, with a max of seven treated states in any given cohort, which leads to imprecision in the estimates. This suggests that perhaps another data avenue may be more fruitful for this question, such as survey data on youth sports participation at a more granular level, which could provide more insight into the effects of the WNBA on youth sports participation. As of now, the imprecision in estimation appears to be a fundamental constraint of the data and how it implies treatment assignment.

Future work would surely need to explore other data avenues, with more granular data on youth sports participation, to further investigate the effects of the WNBA on youth sports participation. If those data avenues are available, the methods posed in my paper would be interesting to apply and extend. For instance, I am massively interested in looking into how metrics of success in the WNBA, like league standings or star player performance, affect youth sports participation, which would be a more direct test of the role model effect. I am also interested in looking at the effects of college basketball success on youth sports participation, which would be interesting to compare to the effects of the WNBA, as, in some regions, college basketball is more popular and has more visibility than the WNBA.

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Tables and Figures

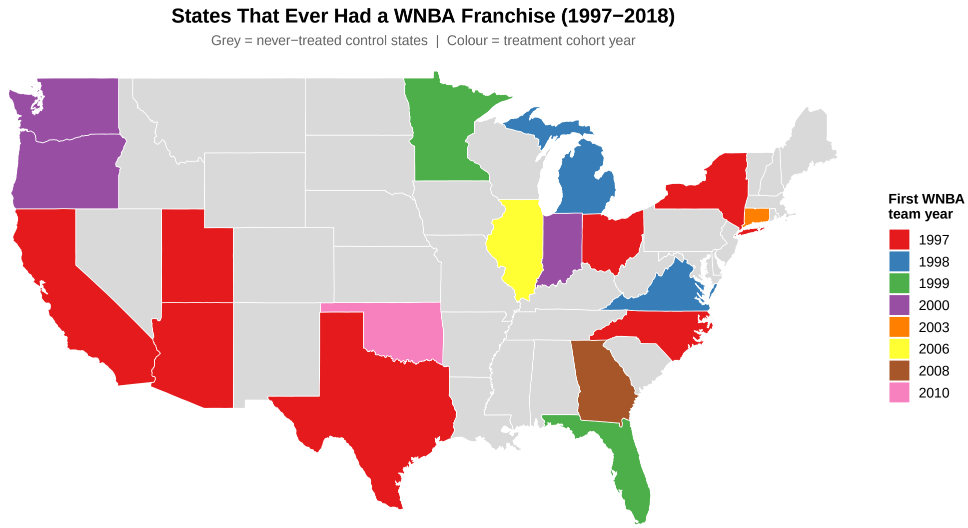

Figure 1: WNBA Franchise Locations by State and Entry Year

Table 1: Summary Statistics: Girls' Sport Participation Rate by Treatment Status

	Never-Treated		Ever-Treated		
Sport	Mean	SD	Mean	SD	Difference
Basketball	21.30	10.68	15.31	6.33	−5.99
Cross Country	6.88	3.20	6.60	2.82	−0.28
Soccer	12.20	6.97	10.90	3.47	−1.30
Track & Field	21.27	9.63	17.19	6.87	−4.08

Note: Means and standard deviations pooled across all state-years. Participation measured as participants per 10,000 total state population (Census-interpolated). Ever-treated = states that ever hosted a WNBA franchise.

Table 2: Mean Girls' Sport Participation Rate by Treatment Cohort

Cohort	N	Basketball	Cross Country	Soccer	Track & Field
1997	7	14.27	6.54	10.17	16.77
1998	2	15.02	6.49	11.15	18.70
1999	2	15.22	6.53	10.68	16.73
2000	3	17.23	7.13	12.53	17.53
2003	1	12.64	7.80	16.08	28.66
2006	1	15.62	6.91	11.90	15.41
2008	1	11.25	5.37	8.97	11.00
2010	1	24.15	5.47	6.84	13.47
Never-Treated	**33**	**21.30**	**6.88**	**12.20**	**21.27**

Note: Cell entries are mean participation rates (per 10,000 total state population) averaged across all state-years in each cohort. Cohort = year of first WNBA franchise in the state.

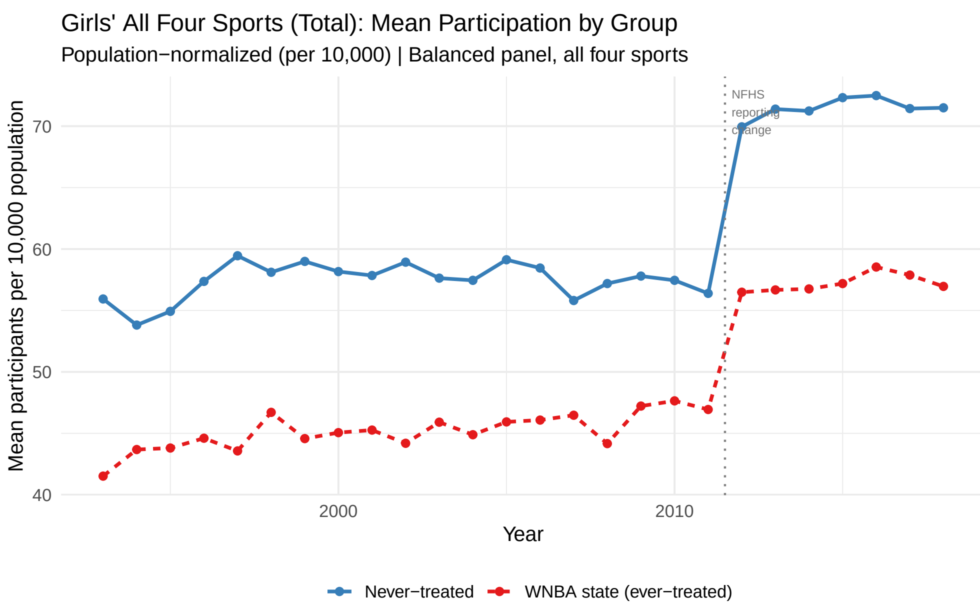

Figure 2: Girls' Mean Participation by Group

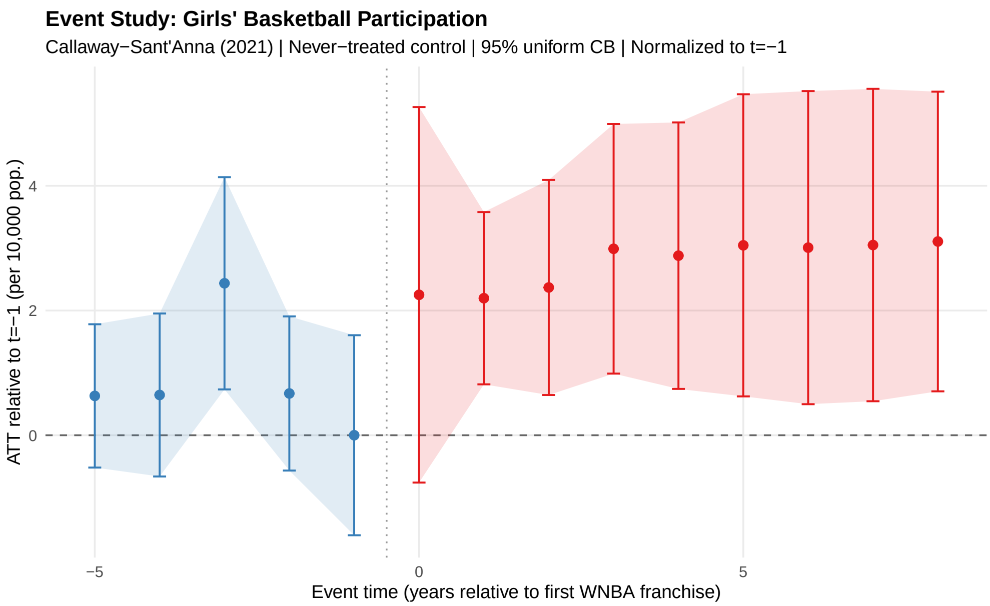

Figure 3: Event Study: Girls' Basketball Participation

Table 3: ATT Estimates Across Estimators and Sports

Sport	TWFE	Callaway-Sant'Anna	Synthetic DiD
Basketball	2.100 (0.900)	2.200 (1.200)	2.100 (0.800)
Soccer	0.000 (0.700)	−0.200 (0.800)	−0.600 (0.500)
Cross Country	0.100 (0.400)	−0.100 (0.500)	−0.300 (0.400)
Track & Field	0.500 (0.800)	0.600 (1.000)	0.300 (0.800)

Note: Standard errors in parentheses. ATT = average treatment effect on the treated, measured in participants per 10,000 total state population. TWFE = two-way fixed effects estimator; Callaway-Sant'Anna = staggered DiD with never-treated control group [Callaway and Sant'Anna, 2020]; Synthetic DiD = Arkhangelsky et al. [2021], stacked by cohort.

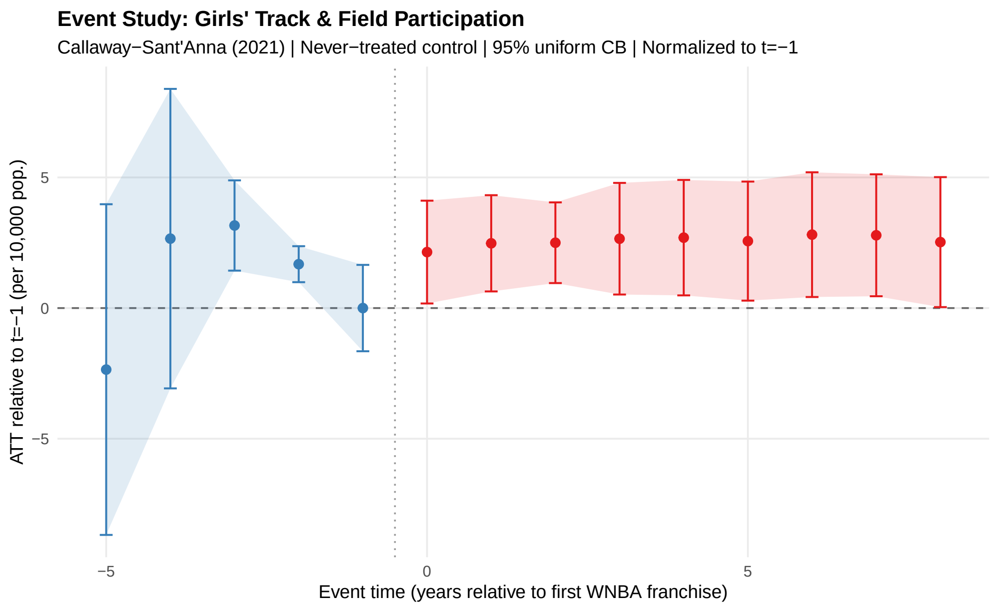

Figure 4: Event Study: Girls' Track & Field Participation

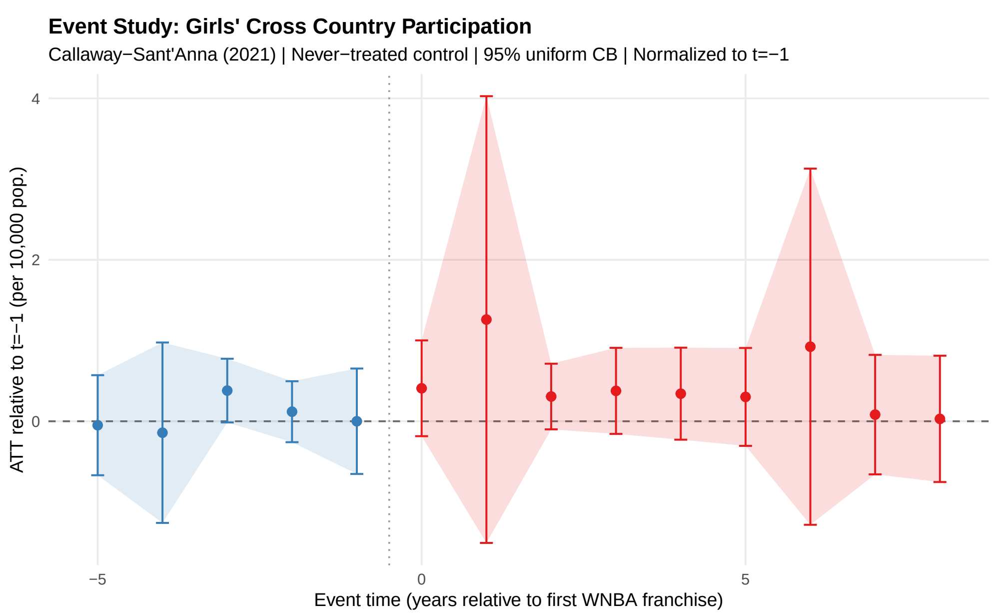

Figure 5: Event Study: Girls' Cross Country Participation

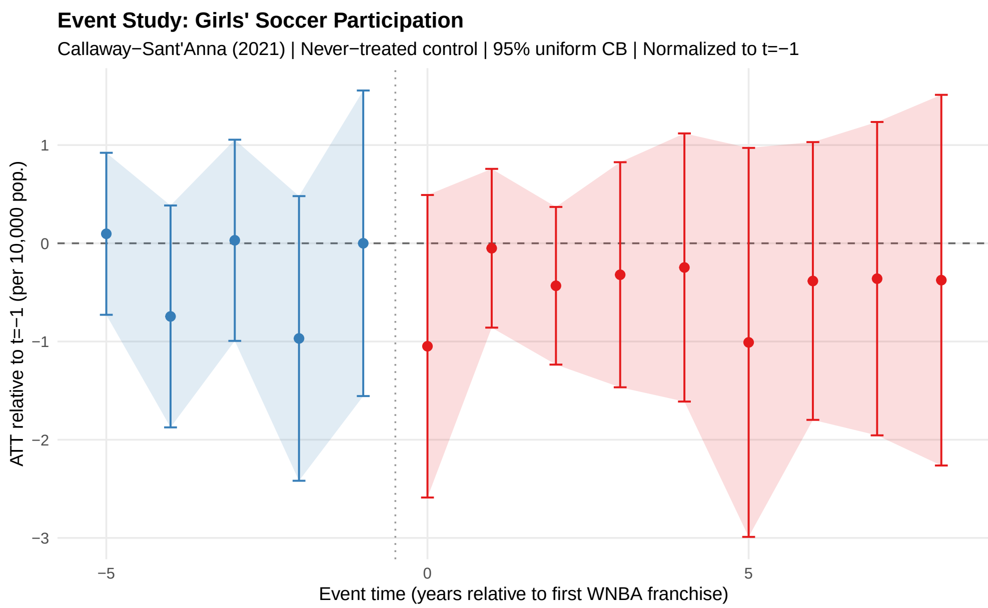

Figure 6: Event Study: Girls' Soccer Participation

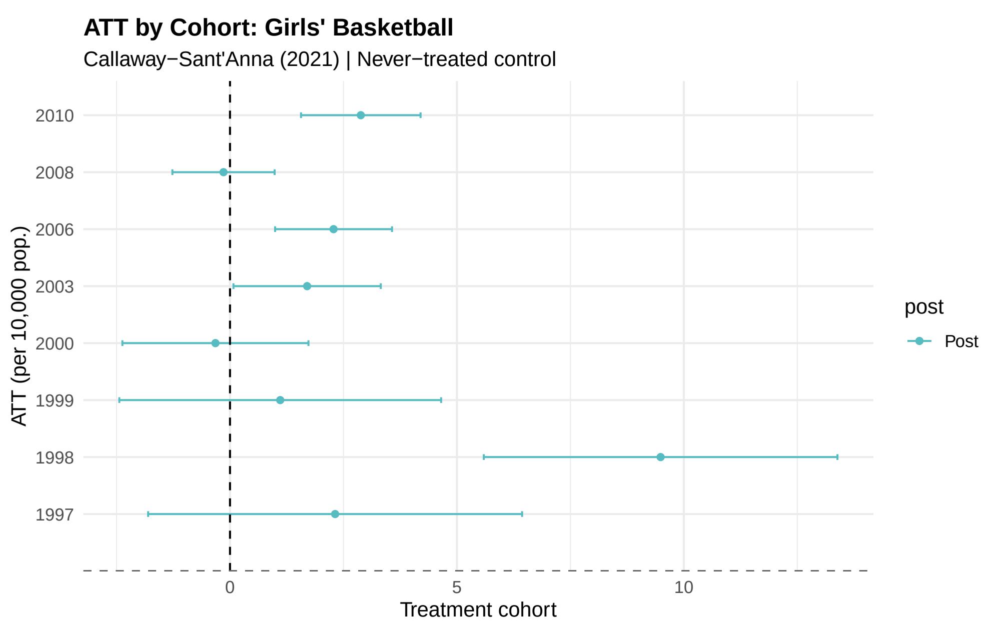

Figure 7: ATT by Cohort: Girls' Basketball

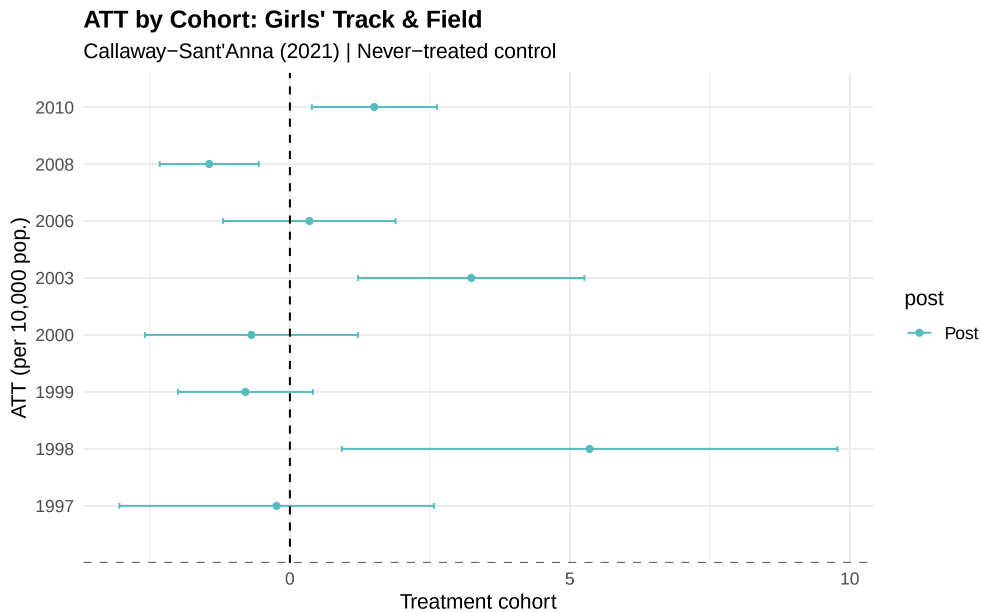

Figure 8: ATT by Cohort: Girls' Track & Field

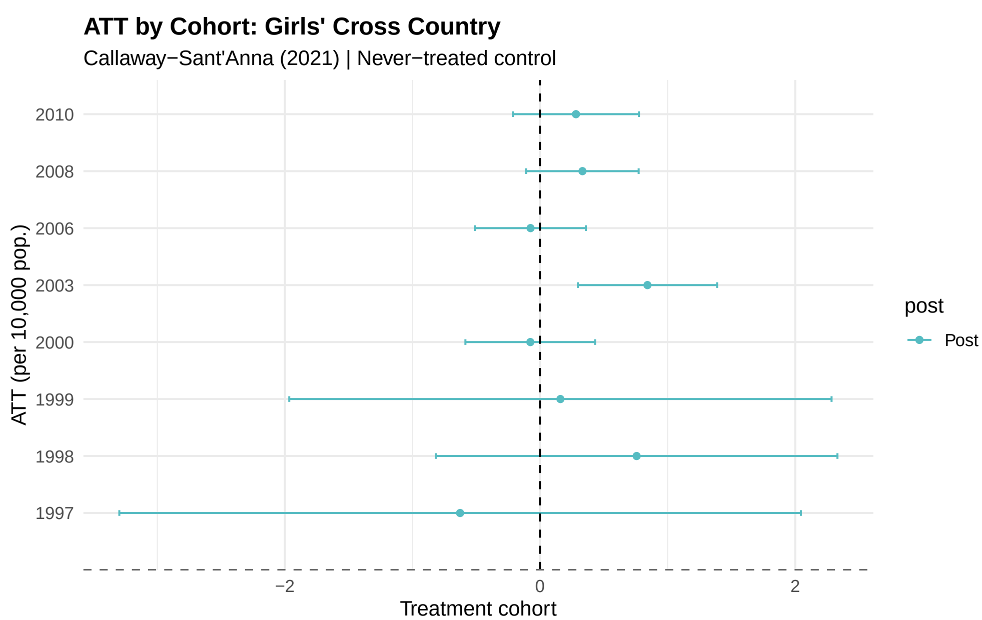

Figure 9: ATT by Cohort: Girls' Cross Country

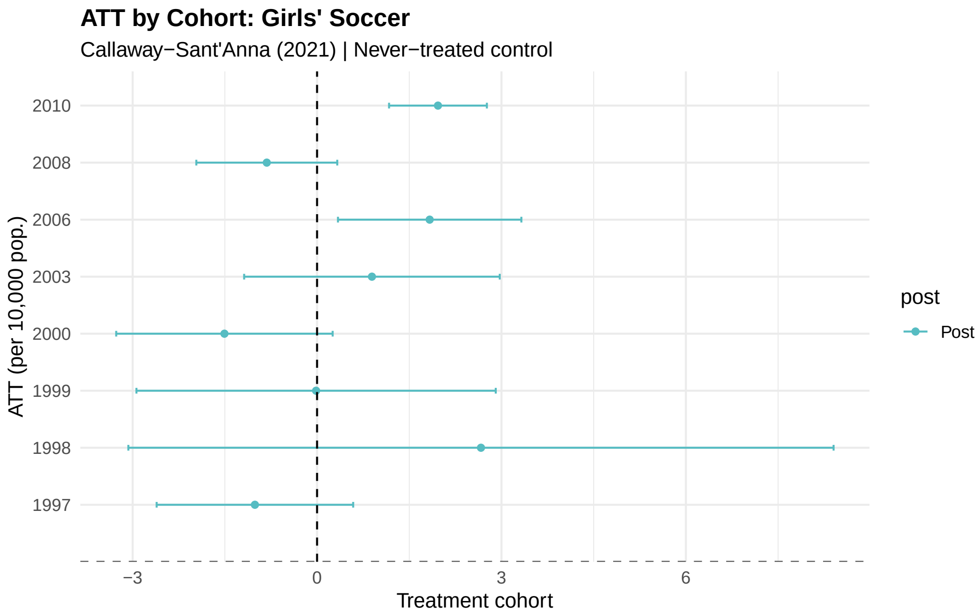

Figure 10: ATT by Cohort: Girls' Soccer

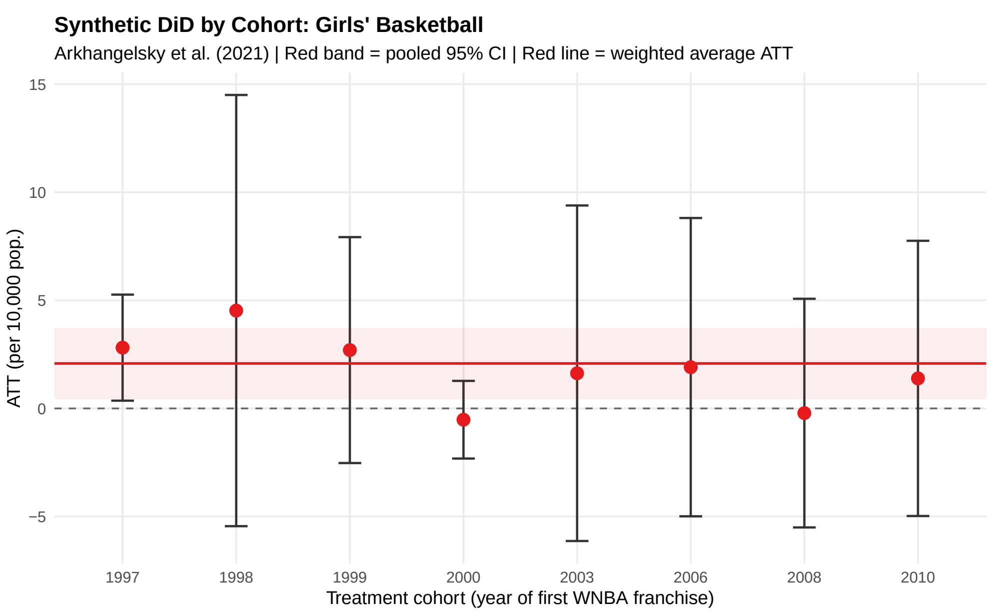

Figure 11: Synthetic DiD by Cohort: Girls' Basketball

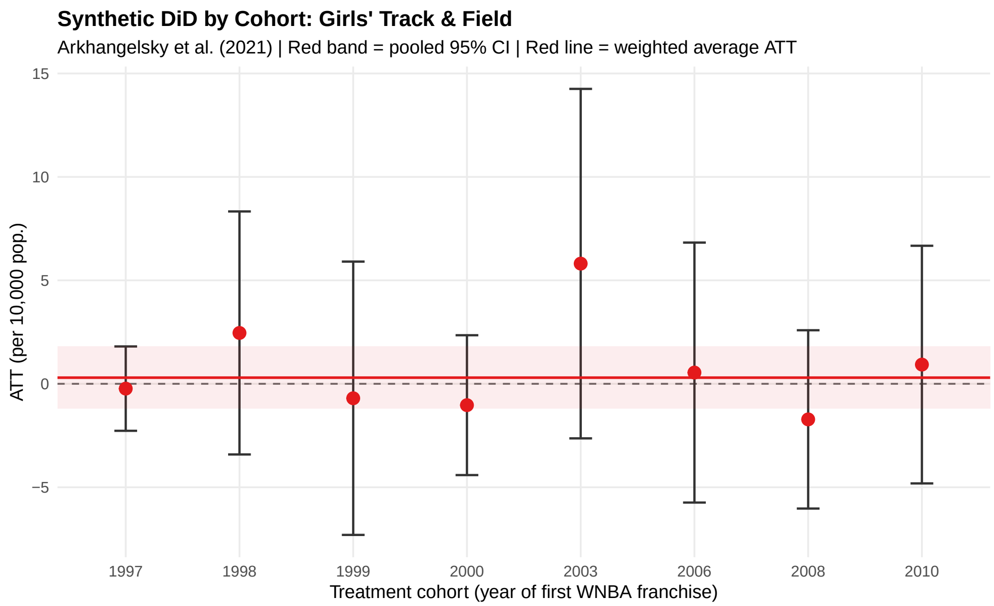

Figure 12: Synthetic DiD by Cohort: Girls' Track & Field

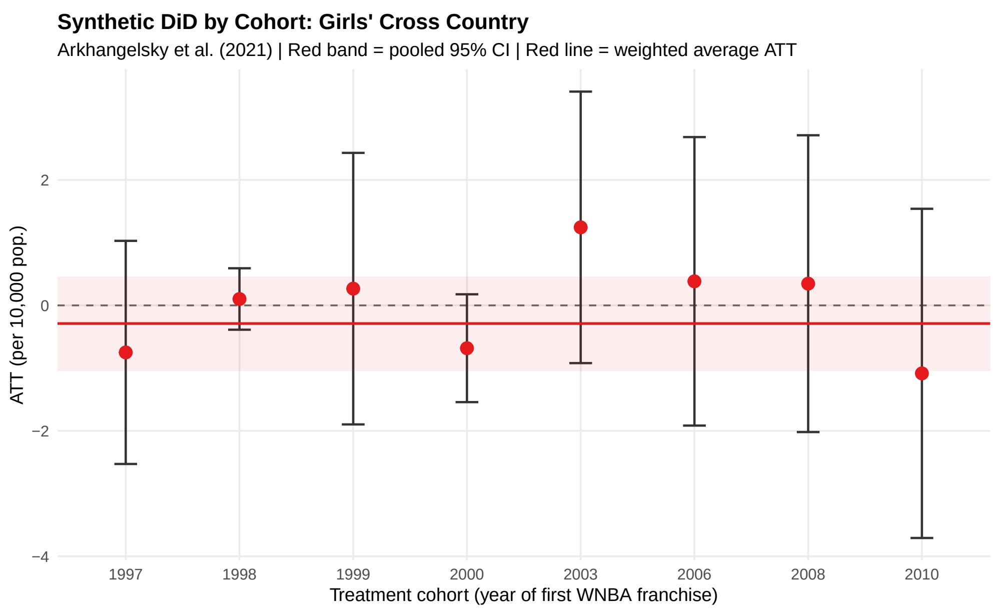

Figure 13: Synthetic DiD by Cohort: Girls' Cross Country

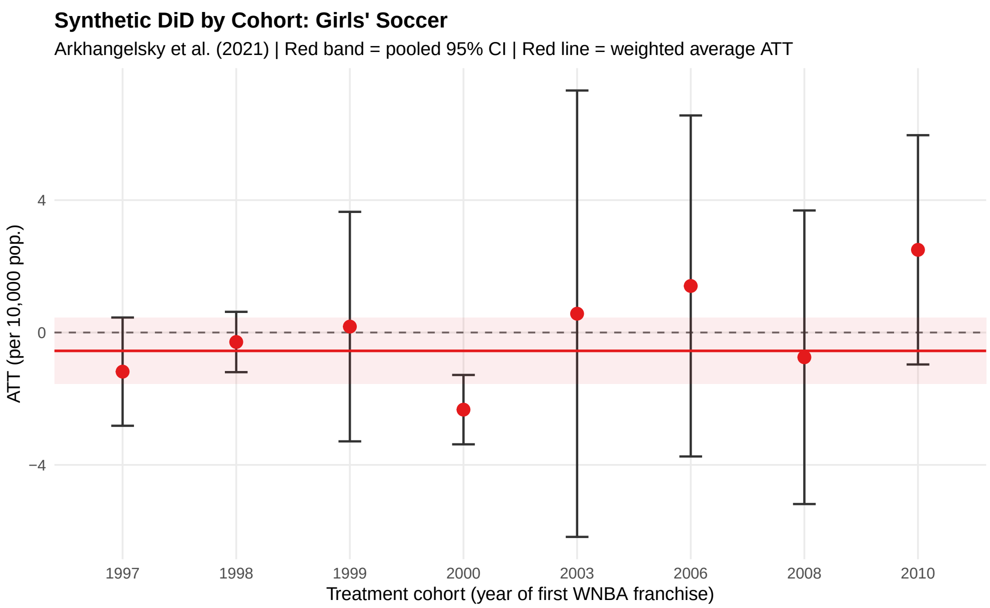

Figure 14: Synthetic DiD by Cohort: Girls' Soccer

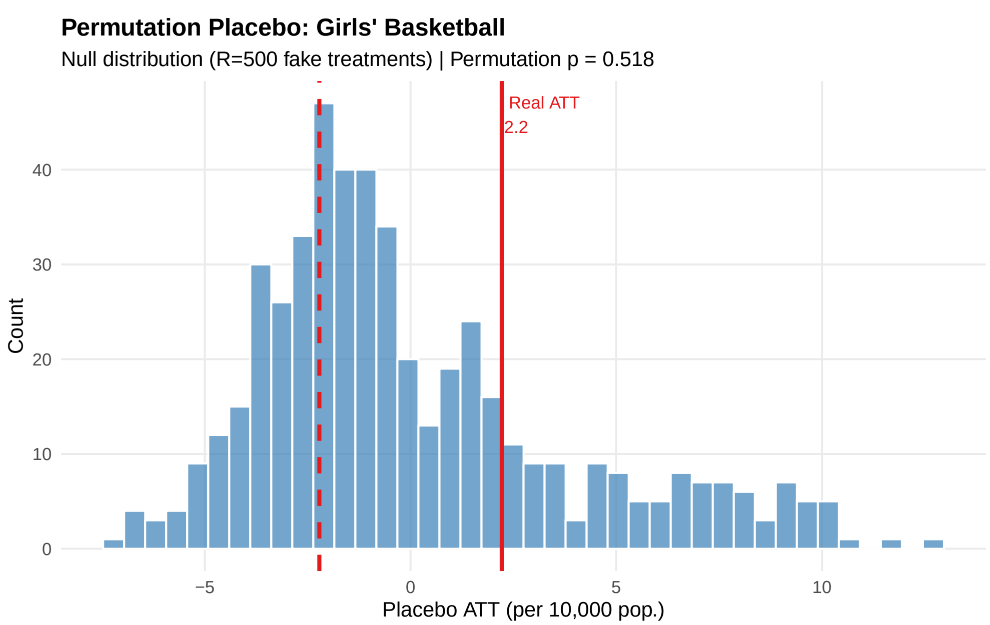

Figure 15: Permutation Placebo Test: Basketball

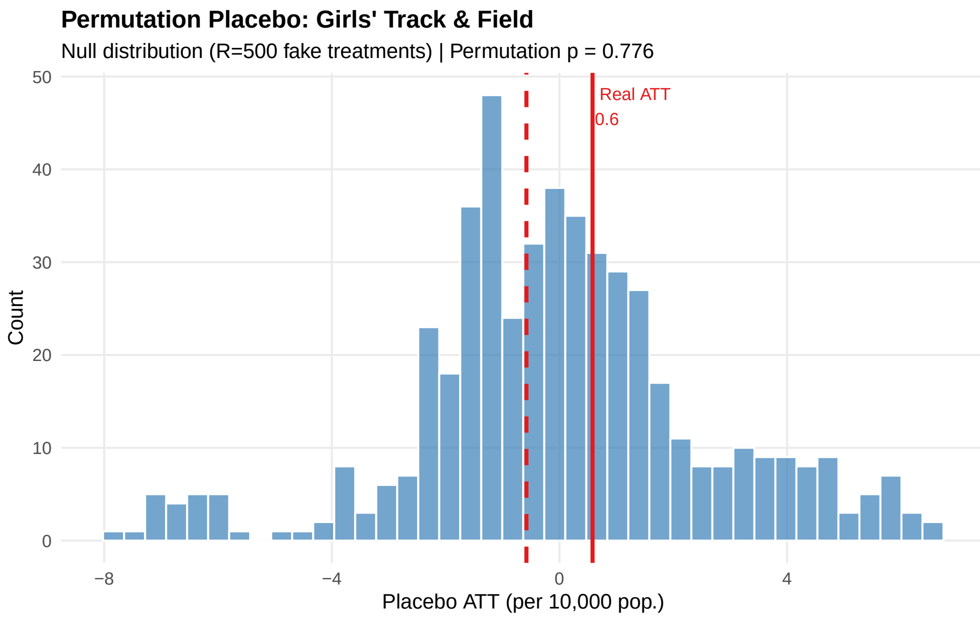

Figure 16: Permutation Placebo Test: Track & Field

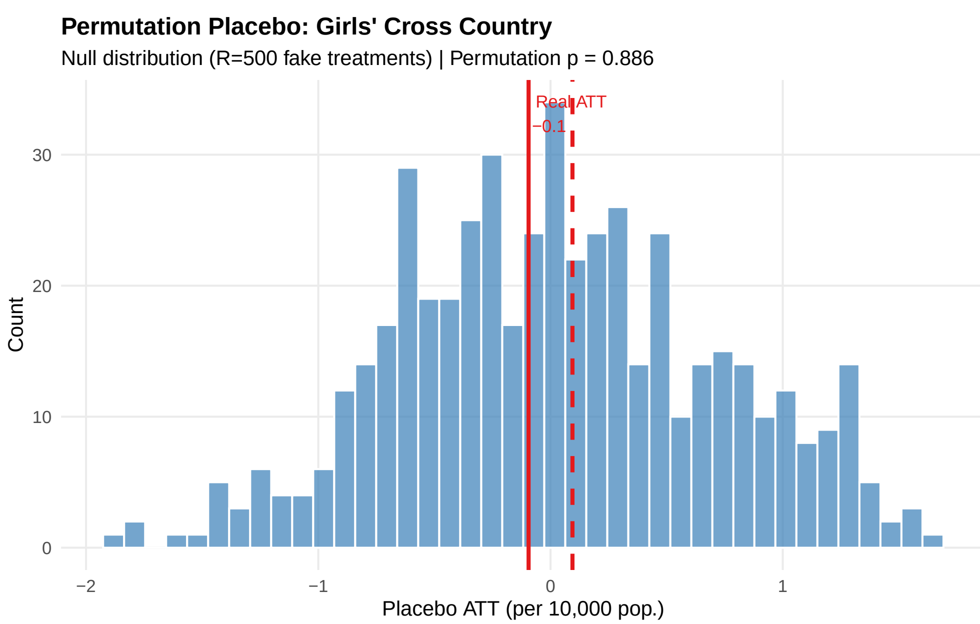

Figure 17: Permutation Placebo Test: Cross Country

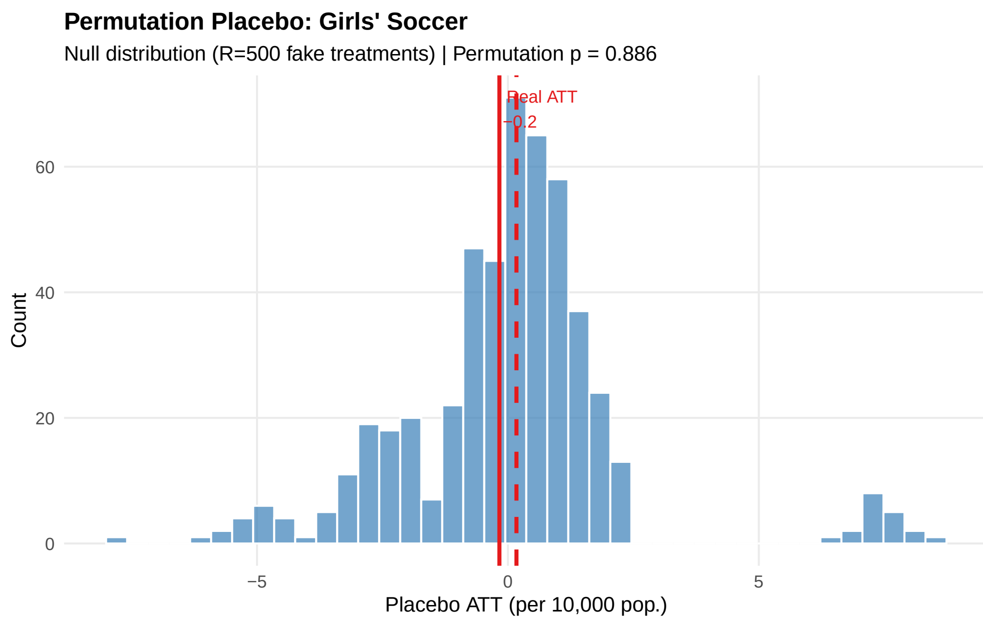

Figure 18: Permutation Placebo Test: Soccer

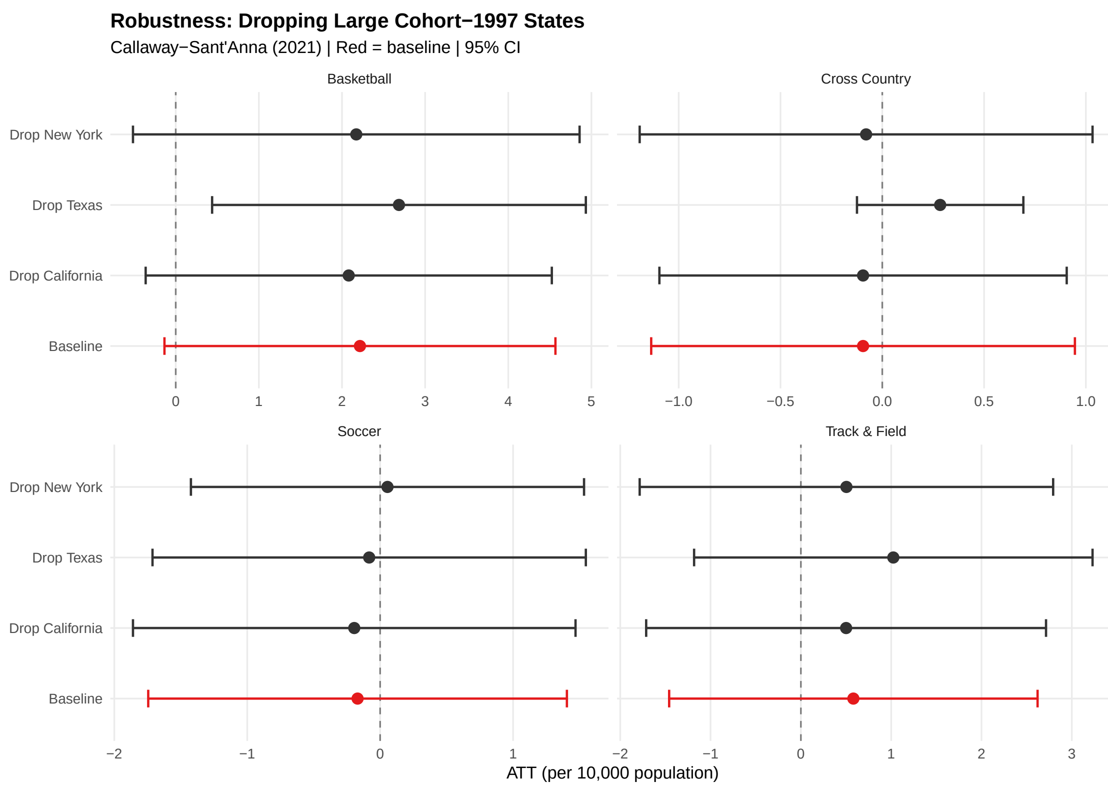

Figure 19: Robustness: Callaway-Sant'Anna ATT After Dropping Large Cohort-1997 States